

Lab 1 – GradPace Descriptive Paper

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1 Introduction

The transition from academic life to professional employment poses a persistent challenge for college students. Despite widespread higher education enrollment, a significant gap remains between the skills graduates possess and those employers seek. The National Association of Colleges and Employers (NACE, 2023) reports that fewer than half of hiring managers consider new graduates fully prepared for entry-level roles, while the World Economic Forum (2023) projects that over half of current job roles will require substantially different skill sets by 2027. This disconnect produces broader consequences — underemployment, misallocated educational investment, and talent shortages in high-demand industries.

Existing career readiness tools address this problem only partially. University career centers offer advising and job fairs, but these services are episodic and lack the personalization needed to target each student’s specific gaps. Platforms like LinkedIn and Handshake surface job listings but provide little structured guidance on skill development. Neither connects a student’s academic record to a quantified readiness metric benchmarked against real industry expectations, nor do they surface local opportunities — such as hackathons and workshops — to help students close identified gaps.

An effective solution must continuously aggregate employer skill requirements into a role-specific benchmark, measure a student’s profile against that benchmark with a quantified score, and connect students to nearby opportunities aligned to their gaps, all while adapting as both the student’s profile and industry demands evolve. GradPace is a career readiness platform built to deliver exactly this. Serving students, alumni, and industry professionals, GradPace combines rule-based scoring with a lightweight machine learning layer for skill extraction and semantic

matching, transforming career readiness from an abstract aspiration into a measurable, actionable roadmap.

2 Product Description

To be completed.

2.1 Key Product Features and Capabilities

To be completed.

2.2 Major Components (Hardware/Software)

To be completed.

3 Identification of Case Study

To be completed.

4 Glossary

API (Application Programming Interface): A set of rules that allow different software applications to communicate and exchange data with each other.

Authentication: The process of verifying a user's identity before allowing access to a system.

Backend: The server-side part of a website or application that handles logic, databases, and data processing.

Benchmark: A role-specific profile of the skills, experiences, certifications, and networking milestones that industry sources collectively indicate are required for a given career track. In GradPace, benchmarks are aggregated dynamically from job postings, public labor datasets, alumni surveys, and employer-submitted data.

Career Readiness: A measure of how prepared a student is to enter a specific career based on skills, experience, and qualifications.

Career Readiness Score: A quantified, 0–100 measure of how closely a user's current profile aligns with the role-specific benchmark, broken down by weighted competency dimensions including technical skills, relevant experience, professional certifications, and networking depth.

Career Roadmap: A structured, visual plan that outlines a student's progress, goals, and required steps toward a specific career path.

Cloud Computing: The use of remote servers on the internet to store data and run applications instead of a local computer.

Data Leak: The accidental or unauthorized exposure of sensitive or private information.

Database: An organized collection of data that can be stored, accessed, and managed electronically.

Encryption: A security method that converts data into a coded form so that only authorized parties can read it.

Frontend: The user-facing part of a website or application that users interact with in their browser.

Gap Detection (Early-Warning System): A feature that identifies missing skills, experiences, or qualifications needed for a target career.

Industry Readiness Benchmarking: The process of comparing a student's profile against expected skills and qualifications for specific careers.

Location-Based Event Recommendations: Suggestions for nearby internships, networking events, or career opportunities based on user location.

Professional Data Input: User-provided information such as coursework, projects, internships, and certifications used to build a profile.

Skill Gap: A competency area in which a user's current profile falls below the benchmark threshold for their target career role. Skill gaps drive GradPace's opportunity recommendations.

Survey Results Dashboard: A feature that displays aggregated insights from alumni data to inform students and administrators.

System Administrator: A role responsible for managing the platform, maintaining data quality, and overseeing system functionality.

Token (AI): A small unit of text processed by an AI model, such as a word, part of a word, or punctuation.

User Interface (UI): The visual elements of a software application that allow users to interact with it.

Web Scraping: The automated process of extracting structured data from web pages. GradPace uses web scraping to collect local professional events, workshops, hackathons, and certification opportunities from publicly accessible community and university event sources.

Web Server: A computer system or software that stores websites and delivers web pages to users when they request them through a browser.

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